Universal Owner's Guide

Comprehensive Pool & Spa Care for Melbourne Homeowners



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Melbourne Pools, Made Easy • #1 Rated Pool Service in the City of Yarra

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Welcome to Pool Ownership

Congratulations on your swimming pool! Whether you have just built a new pool, moved into a property with an existing pool, or are simply looking to improve your maintenance routine, this guide is designed to provide you with the essential knowledge needed to keep your pool pristine, safe, and ready to enjoy all year round.



At **Palatial Pool Services**, we believe that pool ownership should be a joy, not a burden. We are proud to be recognised as the **Best Pool Service in the City of Yarra**, delivering highly intelligent, capable, and pragmatic pool services across Melbourne. With a 10-year workmanship guarantee and a team of licensed professionals, we stand behind every job we complete.

The Palatial Philosophy

We utilise science and serious technology to take pool care to the next level. From advanced acoustic leak detection and thermal imaging to precise water chemistry balancing with calibrated digital instruments, our approach ensures your pool operates at peak efficiency while protecting your valuable equipment investment.

The Four Pillars of Pool Care

Successful pool maintenance relies on four fundamental pillars working together in harmony. Neglecting any one of these will inevitably lead to water quality issues, equipment damage, or both.

1. **Circulation:** Moving water is healthy water. Proper circulation ensures chemicals are distributed evenly and water passes through the filtration system. Stagnant water breeds algae and bacteria.
2. **Filtration:** The physical removal of debris and microscopic particles from the water. Your filter is the kidney of the pool system.
3. **Cleaning:** Brushing walls, vacuuming the floor, and emptying baskets to remove organic matter before it breaks down and consumes your sanitiser.
4. **Chemistry:** Maintaining the correct balance of sanitiser, pH, and other chemical parameters to ensure water is safe for swimmers, comfortable on the skin, and non-corrosive to equipment.

Need Professional Help?

Palatial Pool Services offers comprehensive maintenance plans, one-off diagnostic visits, and onsite equipment lessons by our qualified Trainer and Assessor (TAE10). We can teach you about all aspects related to your individual pool needs, including chemistry, equipment settings, and all your responsibilities as a pool owner. We are just a phone call away: **0452 157 694**.

Understanding Your Equipment

Your pool's equipment pad is the heart of the swimming pool system. Understanding how water flows through this equipment is crucial for effective maintenance, troubleshooting, and communicating issues to your service technician.

The Circulation Path

Water follows a specific path through your equipment. It is drawn from the pool via suction, pushed through the cleaning and heating systems under pressure, and returned to the pool through wall-mounted return jets.



> **The Standard Flow Sequence**
>
> **Skimmer Box / Main Drain → Pool Pump → Media or Cartridge Filter → Heater (if fitted) → Chlorinator / Sanitiser → Return Jets**
>
> Every component after the pump is under positive pressure. Every component before the pump is under negative (suction) pressure. This is why air leaks on the suction side cause the pump to lose prime.

Core Components

1. The Pool Pump

The pump is the engine of the circulation system. It draws water from the skimmer box and main drain, then pushes it through the rest of the equipment. Modern variable-speed pumps (like the **Neptune X-Pro Full Inverter** from Pool Pro Australia) offer significant energy savings by running at lower speeds for longer periods, reducing noise and electricity costs by up to 80%.

2. The Filter

The filter physically removes dirt, debris, and microscopic particles from the water. The most common types in residential pools are **Media Filters** (using sand or glass media) and **Cartridge Filters**. Media filters require regular "backwashing" to flush trapped debris out of the media bed. Cartridge filters require periodic removal and hosing.

3. The Heater (Optional)

If installed, the heater warms the water after it has been filtered but before it is sanitised. Common types include **Inverter Heat Pumps** (most efficient for regular use), **Gas Heaters** (fastest heating for spas and occasional use), and **Solar Heating** systems (lowest running cost but weather-dependent).

4. The Sanitisation System

This is the final stage before water returns to the pool. **Saltwater chlorinators** (like the Pool Pro MineralX) convert dissolved salt into chlorine gas through electrolysis. **Liquid chemical feeders** (like the Neptune pH Controller) inject precise amounts of liquid chlorine and acid into the water stream automatically. Manual dosing is also common in many residential setups.

> **Equipment Tip: Know Your Pressure Gauge**
>
> The pressure gauge on top of your filter is one of the most important diagnostic tools you have. Note the "clean starting pressure" after a fresh backwash (typically 50–80 kPa). When the pressure rises 50–70 kPa above that baseline, it is time to backwash. A sudden drop to zero may indicate a broken gauge or a pump issue.

Filtration & Pump Runtimes

Proper filtration is arguably the most important aspect of pool maintenance. Even with perfect chemistry, a pool without adequate filtration will quickly become cloudy, develop algae, and become unsafe for swimming. The pump must run long enough each day to turn over the entire pool volume at least once.



Pump Runtimes: The Single Continuous Block Rule

CRITICAL: Do Not Split Your Pump Runtimes

Palatial Pools **strongly advises against splitting your pump runtimes** into multiple short blocks throughout the day. **Always run your pump as a single continuous block.**

Splitting runtimes means the pump is turned off during peak UV and heat hours — typically the middle of the day. This is precisely when chlorine demand is highest, UV degradation is most intense, and bacterial growth is most rapid. A continuous run ensures constant sanitiser production and uninterrupted circulation during the hours your pool needs it most.

Run the pump as long as needed to keep the pool clear — up to 10 hours in some cases.

Recommended Continuous Runtimes

Season	Daily Runtime	Notes
Summer (Peak)	8–10 hours	Higher temperatures and UV demand longer runs. Heavy use periods may require even more.
Spring / Autumn	6–8 hours	Moderate demand. Adjust based on water clarity and chlorine consumption.
Winter (Off-Season)	4–6 hours	Lower demand, but never turn the pump off entirely. Stagnant water breeds problems.

Setting Your Timer

Set your pump timer to start in the early morning (around 6–7 am) so the pump is running through the hottest part of the day when chlorine demand is highest. If your pool requires 8 hours of runtime, a 6 am start means the pump runs until 2 pm, covering the peak UV period. Adjust seasonally — increase runtime as temperatures rise and reduce it as they fall, but always maintain a minimum of 4 hours even in the depths of winter.

Why Continuous Matters: The Science

Chlorine is constantly being consumed by UV light, organic matter, and bather waste. When the pump stops, chlorine production stops (in salt chlorinator systems), circulation stops, and the water in the pipes and filter becomes stagnant. Meanwhile, the sun continues to destroy the chlorine that remains in the pool. By the time the pump restarts hours later, the pool may have lost its sanitiser residual entirely, creating a window for algae and bacteria to establish.

Filter Types Comparison

Feature	Sand / Glass Media	Cartridge
Filtration Fineness	20–40 microns (sand) / 5–15 microns (glass)	10–20 microns
Cleaning Method	Backwash via multiport valve	Remove and hose down
Media Lifespan	5–7 years (sand) / 8–12 years (glass)	1–3 years per cartridge
Water Waste	Uses pool water during backwash	Minimal (hose water only)
Best For	Most residential pools	Smaller pools, water-saving setups

Pro Tip: Glass Media Upgrade

If your sand filter is due for a media change, Palatial Pools recommends upgrading to **recycled glass media**. Glass provides superior filtration down to finer micron levels, resists bacterial channelling, requires less frequent backwashing, and lasts significantly longer than traditional sand. Sand changes start from $775 including labour and materials.

Source: Filtration standards referenced from SPASA Victoria technical guidelines.

Media Filter Backwashing Guide

Backwashing reverses the flow of water through the filter media, lifting the trapped dirt and flushing it out through the waste line. This process is essential for maintaining water clarity and preventing equipment damage from excessive pressure buildup inside the filter tank.

When to Backwash

Monitor the pressure gauge on top of your filter tank. Note the **clean baseline pressure** after a fresh backwash (typically 50–100 kPa depending on your system). When the pressure rises **50–70 kPa above baseline**, it is time to backwash. Do not backwash on a fixed schedule — backwash only when the pressure tells you to. Over-backwashing wastes water and removes beneficial filter bed compaction that actually improves filtration. Under-backwashing restricts flow and strains the pump.

Golden Rule of Multiport Valves

NEVER turn the multiport valve handle while the pump is running. Always turn the pump OFF before changing the valve position. Failing to do so will blow the internal spider gasket, causing water to bypass the filter and potentially flood the equipment area. This is a common and expensive mistake.



Step-by-Step Backwash Procedure

1. **Turn the pump OFF.**
2. Press down firmly on the multiport valve handle and rotate it to the **BACKWASH** position.
3. Ensure the waste line is open or the backwash hose is rolled out to an appropriate drainage area (not into the stormwater drain).
4. **Turn the pump ON.**
5. Watch the sight glass on the side of the valve. The water will initially run dirty, then gradually clear. This usually takes **2 to 3 minutes**.
6. Once the water in the sight glass runs completely clear, **turn the pump OFF.**
7. Rotate the handle to the **RINSE** position.
8. **Turn the pump ON** for 15–30 seconds. This settles the media bed and flushes remaining dirty water out the waste line instead of back into the pool.
9. **Turn the pump OFF.**
10. Rotate the handle back to the **FILTER** position.
11. **Turn the pump ON** to resume normal filtration.

Multiport Valve Positions

Position	Function
Filter	Normal operation. Water flows through the media and returns to the pool.
Backwash	Reverses flow to flush trapped dirt out the waste line.
Rinse	Settles media bed after backwash; sends water to waste briefly.
Waste	Bypasses the filter entirely and sends water straight to waste. Used for vacuuming heavy debris or lowering water level.
Recirculate	Bypasses the filter but returns water to the pool. Used when the filter is broken or during certain chemical treatments.
Closed	Shuts off all flow. Used for pump maintenance or winterising.

> **Post-Backwash: Check Your Water Level**
>
> Backwashing removes a significant amount of water from the pool. **Always check the water level after backwashing.** The water should sit halfway up the skimmer box opening. If it is too low, the pump will draw in air, lose prime, and potentially run dry — causing severe and expensive damage to the pump seal and impeller. Top up with a garden hose immediately if required. After topping up, retest your water chemistry as the fresh tap water will have diluted your chemical balance.

Water Chemistry Basics

Maintaining correct water chemistry is vital for swimmer health, water clarity, and the longevity of your pool surface and equipment. Unbalanced water can become **corrosive** (destroying heaters, pump seals, and metal fittings) or **scale-forming** (calcifying salt cells, clogging pipes, and roughening surfaces). Understanding the key parameters and their interactions is the foundation of effective pool care.

Ideal Chemistry Targets

Parameter	Ideal Range	Palatial Recommendation
Free Chlorine	2.0–4.0 ppm	**Target approximately 4 ppm** as a working level. This provides a buffer during hot weather and heavy bather loads, ensuring the pool never drops below the safe minimum.
pH Level	7.2–7.6	**7.2–7.4 ideal.** This range maximises chlorine's disinfection power (see next section).
Total Alkalinity	80–120 ppm	**100 ppm.** TA acts as a pH buffer, preventing rapid pH fluctuations.
Cyanuric Acid (Stabiliser)	30–50 ppm	**40 ppm.** Protects chlorine from UV degradation. Too high causes "chlorine lock."
Calcium Hardness	150–200 ppm (Concrete) 125+ ppm (Fibreglass/ Vinyl)	Depends on pool surface. Concrete/pebble pools need higher calcium to prevent the water from leaching calcium from the shell.
Salt Level	3,000–5,000 ppm	Check your specific chlorinator manual. The Pool Pro MineralX operates at ultra-low salt levels.

Why Palatial Pools Recommends Approximately 4 ppm Free Chlorine

The standard Australian guideline is 2–4 ppm for residential pools. However, in practice, chlorine is constantly being consumed by UV light, bather waste, and organic debris. If you target the middle of the range (3 ppm), a hot afternoon with heavy use can easily drop you below 2 ppm — the minimum safe level. By targeting **approximately 4 ppm**, you build in a safety buffer that keeps the pool protected even during peak demand periods. This is especially important in Melbourne's hot summers.

Understanding Chlorine Lock

If Cyanuric Acid (stabiliser) levels rise above 80–100 ppm, it begins to "lock" the chlorine, preventing it from effectively sanitising the water. The test kit will show adequate free chlorine, but the pool will still develop algae. The only way to reduce stabiliser is to partially drain and refill the pool with fresh water. Avoid using stabilised chlorine products (dichlor/trichlor tablets) as they continuously add stabiliser with every dose.

Water Balance Quick Reference

Use this guide to quickly identify what to adjust and which direction to go.

If...	Then Add...	Product
pH too HIGH (>7.6)	Acid (to lower pH)	Hydrochloric Acid (HCl)
pH too LOW (<7.2)	Alkali (to raise pH)	Sodium Carbonate (Soda Ash)
Chlorine too LOW (<2 ppm)	Chlorine (shock dose)	Liquid Pool Chlorine (unstabilised)
Alkalinity too LOW (<80 ppm)	Buffer (to raise TA)	Sodium Bicarbonate (Buffer)
Calcium too LOW	Calcium (to raise hardness)	Calcium Chloride
Stabiliser too LOW (<30 ppm)	Stabiliser	Cyanuric Acid

Always adjust pH first before correcting other parameters. pH affects the accuracy of all other readings.

Source: Chemistry parameters align with SPASA Victoria guidelines and Victorian Department of Health recommendations for residential swimming pools.

pH & Chlorine Effectiveness

The relationship between pH and chlorine effectiveness is one of the most critical — and most misunderstood — aspects of pool chemistry. Many pool owners focus on adding more chlorine when their pool develops problems, when in reality, **the pH level determines how effective the chlorine already in the water actually is.**

The Science: Hypochlorous Acid vs. Hypochlorite Ion

When chlorine dissolves in water, it forms two chemical species: **Hypochlorous Acid (HOCl)** and **Hypochlorite Ion (OCl^-)**. Hypochlorous acid is the **active, powerful disinfectant** — it is approximately 80–100 times more effective at killing bacteria and algae than the hypochlorite ion. The ratio between these two forms is controlled entirely by the pH of the water.

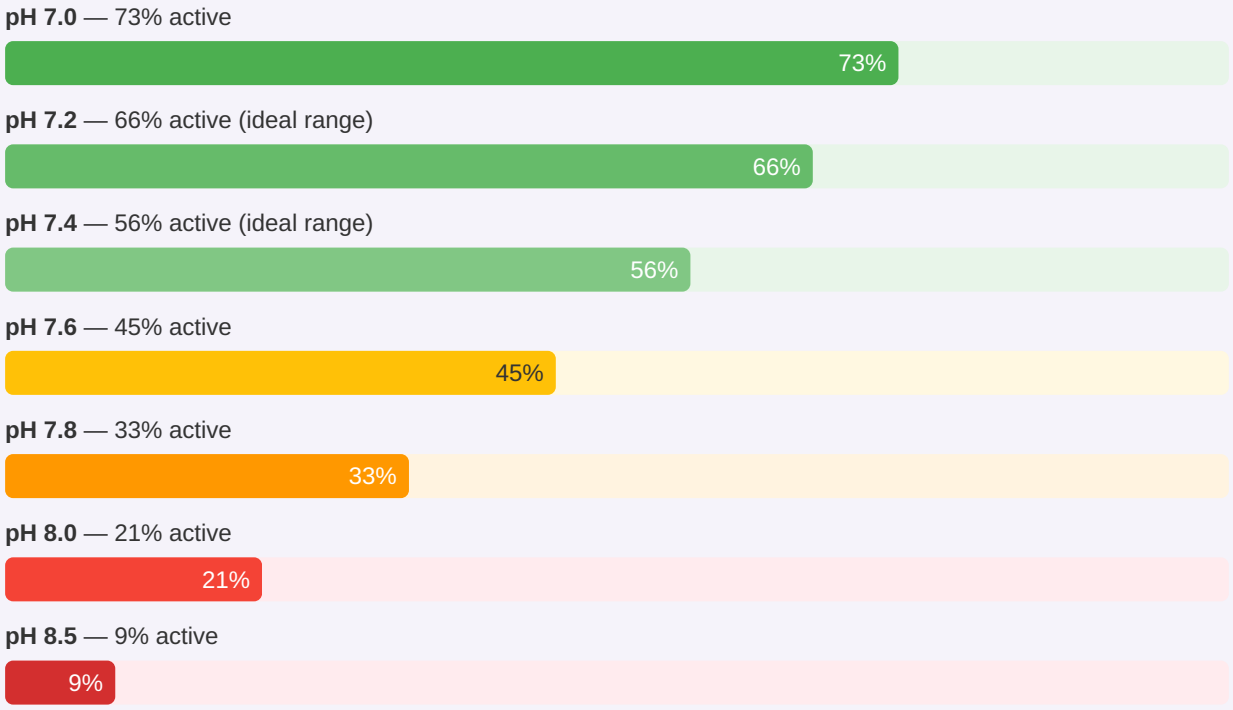
The Key Takeaway

High pH = Weak Chlorine. No matter how much chlorine you add, if the pH is above 7.8, the vast majority of it converts to the ineffective hypochlorite ion form. You are essentially wasting chemicals and leaving your pool vulnerable to algae and bacteria. **Controlling pH is controlling chlorine effectiveness.**

pH vs. Active Chlorine (HOCl) — The Numbers

pH Level	% Active Chlorine (HOCl)	% Inactive (OCl^-)	Effectiveness Rating
7.0	**73%**	**27%**	**★★★★★ Excellent**
7.2	**66%**	**34%**	**★★★★★ Excellent**
7.4	**56%**	**44%**	**★★★★ Very Good**
7.6	**45%**	55%	★★★ Acceptable
7.8	**33%**	67%	★★ Poor
8.0	**21%**	**79%**	**★ Very Poor**
8.5	**9%**	**91%**	**✗ Almost Ineffective**

Visual Guide: Active Chlorine at Each pH Level



Green = ideal range. Yellow = acceptable. Orange/Red = chlorine increasingly ineffective.

Reading the Chart

At the ideal pH of **7.2–7.4**, approximately **56–66%** of your chlorine is in the powerful HOCl form — actively killing bacteria, viruses, and algae on contact. As pH rises to **7.8**, only 33% remains active. At **pH 8.0**, over 80% of your chlorine has converted to the weak hypochlorite ion — meaning you would need to add roughly **three times as much chlorine** to achieve the same disinfection as a pool at pH 7.2. At **pH 8.5+**, chlorine is almost completely ineffective regardless of concentration.

Why This Matters in Practice

High pH is one of the most common causes of algae and bacterial problems, even when chlorine test results appear adequate. A pool owner may test their water, see a free chlorine reading of 3 ppm, and assume the pool is safe. But if the pH has drifted to 8.0, only about 0.6 ppm of that chlorine is actually working as an effective disinfectant — well below the safe minimum.

The Palatial Pools Recommendation

Keep your pH in the **7.2–7.4 range** at all times. This maximises chlorine's disinfection power, reduces chemical consumption (you need less chlorine to achieve the same result), protects swimmer comfort (the pH of human tears is approximately 7.4), and saves you money. Test pH at least **once per week** alongside free chlorine — these are the two most critical parameters for day-to-day pool health.

Common Causes of Rising pH

Salt chlorinators naturally raise pH as a byproduct of the electrolysis process. This means salt pools almost always need regular acid additions. **High total alkalinity** also resists pH changes downward, making it harder to keep pH low. **Aeration** (water features, spa jets, waterfalls) drives CO_2 out of the water, which raises pH. If your pH is constantly rising, check these factors first.

Source: HOCl/OCl^- dissociation data based on established aqueous chemistry research, referenced by SPASA Victoria and the Victorian Department of Health.

Digital Water Testing

Accurate water testing is the foundation of effective pool chemistry management. Without reliable test results, you are essentially dosing chemicals blindly — risking both under-treatment (unsafe water) and over-treatment (wasted money and potential equipment damage).

Why Palatial Pools Recommends Digital Photometers

The Problem with Dip Strips and Liquid Dye Tests

Traditional dip strips and liquid reagent (DPD) colour-matching tests rely entirely on **visual colour interpretation**. The user must compare a coloured sample against a printed colour chart and make a subjective judgement about which shade it matches. The problem is that colour perception varies significantly from person to person — influenced by lighting conditions, colour blindness (affecting up to 8% of men), age-related vision changes, and even the angle at which the sample is viewed.

This means two people testing the same water sample can arrive at meaningfully different readings. In pool chemistry, where the difference between a pH of 7.4 and 7.8 has dramatic implications for chlorine effectiveness (as shown in the previous section), this subjectivity is a serious limitation.

The Digital Advantage

A calibrated **digital photometer** passes a precise beam of light through the water sample and measures the exact wavelength absorption caused by the reagent reaction. It then displays a **numerical reading** — removing all human colour interpretation from the process. Results are repeatable, precise, and not affected by ambient lighting or the operator's eyesight.

How to Take a Proper Water Sample

For accurate results, collect your water sample from **elbow depth** (approximately 30cm below the surface) and away from return jets or the skimmer box. Surface water is exposed to more UV and evaporation, which can give misleadingly low chlorine readings. Water near return jets has just been through the chlorinator and may read artificially high. A mid-depth sample from the centre of the pool gives the most representative reading of overall water conditions.

Testing Frequency Recommendations

Test	Frequency	Method
Free Chlorine	Once per week	Digital photometer or quality liquid test kit
pH	Once per week	Digital photometer or quality liquid test kit
Total Alkalinity	Monthly	Digital photometer (full spectrum test)
Cyanuric Acid (Stabiliser)	Monthly	Digital photometer (full spectrum test)
Calcium Hardness	Monthly	Digital photometer (full spectrum test)
Salt Level	Monthly	Digital salt meter or chlorinator display
Phosphates	Quarterly or if algae persists	Dedicated phosphate test kit

Weekly Home Testing: Keep It Simple

For your regular weekly home testing, focus on the **two most critical and easiest-to-adjust parameters: Free Chlorine and pH**. These are the readings that change most rapidly and have the greatest immediate impact on water safety and clarity. A comprehensive full-spectrum digital test covering Total Alkalinity, Calcium Hardness, Stabiliser, and other parameters is recommended **monthly** — either with your own photometer or by bringing a water sample to your local pool shop.

Investing in a Photometer

Quality digital photometers suitable for residential pool owners are available from Pool Pro Australia and specialist pool equipment suppliers. While the initial investment is higher than a bottle of test strips, the accuracy, consistency, and long-term chemical savings make it worthwhile. Palatial Pools uses professional-grade photometers on every service visit to ensure precise, reliable results for our clients.

Chemical Dosing & Safety

Handling pool chemicals requires respect and strict adherence to safety protocols. Pool chemicals are highly concentrated industrial substances that can cause severe injury, toxic gas release, or dangerous reactions if mishandled or mixed incorrectly.

Recommended Chemicals: Pool Pro Australia

Palatial Pools recommends using high-quality chemicals from **Pool Pro Australia** (www.poolpro.com.au). Their **Neptune range** and **Pool Pro range** offer reliable, professional-grade solutions for all your water balancing needs. Pool Pro is an Australian-owned company with over 30 years of industry experience and more than 3,500 products.

The Absolute Hard Rule

NEVER MIX ACID AND CHLORINE TOGETHER — EVER.

Mixing Hydrochloric Acid and Liquid Chlorine (Sodium Hypochlorite) creates **highly toxic, potentially lethal chlorine gas instantly**. This reaction can occur even from residual splashes in a shared bucket. Always add chemicals to the pool separately, allowing the pump to circulate the water for **at least 30 minutes between different chemical additions**. Use separate, clearly labelled measuring containers for acid and chlorine.

Safe Dosing Procedures

Adding Hydrochloric Acid (To Lower pH)

1. Always wear protective eyewear and chemical-resistant gloves.
2. Ensure the pool pump is running.
3. **Dilution Rule:** Always add ACID to WATER — never water to acid. Fill a clean plastic bucket with pool water first, then carefully measure and pour the required acid into the bucket.
4. Walk around the deep end of the pool, slowly pouring the diluted acid mixture into the water, keeping it away from pool walls, fittings, and the skimmer box.
5. Allow the pump to circulate for at least 2 hours before retesting or swimming.

Adding Pool Pro pH Buffer (To Raise Total Alkalinity)

Pre-dissolve the powder in a bucket of pool water, then broadcast the solution evenly across the surface of the pool while the pump is running. Brushing the pool walls and floor afterward helps distribute the chemical and dissolve any residue that settles.

Adding Pool Pro Starlight Ultrafine Pool Salt

Cut open the bags and pour the salt directly into the shallow end of the pool. Use a pool brush to push the salt around until it is completely dissolved. **Never** pour salt directly into the skimmer box — this sends a massive concentration of salt directly through your equipment, which can damage the chlorinator cell and pump seal.

> **Chemical Storage Safety**
>
> Store all chemicals in a cool, dry, well-ventilated area out of direct sunlight. Keep liquid chlorine and hydrochloric acid stored in **separate areas** — never on the same shelf or in the same cabinet — to prevent accidental mixing in the event of a spill or container failure. Keep all chemicals locked away from children and pets. Dispose of empty containers according to your local council guidelines.

Source: Chemical handling procedures align with WorkSafe Victoria guidelines and Pool Pro Australia Safety Data Sheets (SDS).

Routine Maintenance Schedule

Consistency is the key to a trouble-free pool. Establishing a regular maintenance rhythm prevents small issues from becoming expensive disasters. The schedule below is designed for a typical Melbourne residential pool.

> **The Palatial Approach to Maintenance**
>
> We believe in honest, practical advice that empowers pool owners. The schedule below focuses on what actually matters for long-term pool health, rather than pushing unnecessary chemicals or service calls. Consistency with the basics — circulation, filtration, and weekly testing — will save you significant money and frustration over the life of your pool.

Weekly Tasks

- **Test Chlorine and pH (once per week):** These are the two most critical parameters for day-to-day pool health. They change the fastest and are the easiest and cheapest to adjust. Use a digital photometer or quality liquid test kit. Adjust dosing as necessary to maintain Free Chlorine at approximately 4 ppm and pH at 7.2–7.4.
- **Empty Skimmer Baskets:** Remove leaves and debris to ensure unrestricted water flow to the pump.
- **Empty Pump Basket:** Turn off the pump, close necessary valves, open the pump lid, and clear the basket. Ensure the lid O-ring is clean and lubricated before resealing.
- **Check Water Level:** Ensure water sits halfway up the skimmer box opening.

- **Brush Walls and Steps:** Brushing disrupts biofilm and algae spores before they can take hold, allowing the filter to capture and remove them.
- **Visual Equipment Check:** Do a quick visual and auditory inspection of the equipment pad. Look for leaks, drips, or unusual pump noises.

> **Weekly Testing Note**
>
> For your regular weekly home testing, focus on **Free Chlorine and pH only**. These are the two readings that fluctuate most and have the greatest immediate impact on water safety. A comprehensive full-spectrum digital test (covering Total Alkalinity, Calcium Hardness, Stabiliser, Phosphates, and Salt) is recommended **once per month** — either using your own calibrated photometer or by taking a water sample to a pool professional.

Monthly Tasks

- **Full-Spectrum Water Test:** Test Total Alkalinity, Calcium Hardness, Cyanuric Acid (Stabiliser), and Salt Level using a digital photometer. Adjust as needed.
- **Backwash Filter:** Backwash the media filter (or clean the cartridge) if the pressure gauge indicates it is required (50–70 kPa above clean baseline).
- **Inspect Salt Cell:** Check the chlorinator cell for white calcium scale buildup. If scaled, clean using a mild acid cell cleaning solution as per manufacturer instructions.
- **Clean Skimmer Weir Door:** Ensure the weir door (the flap inside the skimmer box) swings freely. A stuck weir door reduces skimming efficiency.

Quarterly Tasks

- **Test for Phosphates:** High phosphate levels feed algae. Use Pool Pro Pool Emeraldz if levels are elevated.
- **Inspect Equipment Pad:** Check all pipe connections, valves, and unions for slow leaks or weeping.
- **Clean Pool Cover:** If using a solar blanket, hose it down and check for damage.

Annual Tasks

- **Professional Inspection:** Have Palatial Pools conduct a comprehensive system check, including leak detection, equipment calibration, and a full water analysis.
- **Filter Degreasing:** Use a specialised filter cleaner to remove body oils, sunscreen, and organic buildup from the filter media.
- **O-Ring Maintenance:** Inspect and lubricate all rubber O-rings (pump lid, chlorinator cell, filter lid, valves) with a silicone-based pool lubricant. Never use Vaseline or petroleum-based products — they degrade rubber.

Seasonal Pool Care

Melbourne's variable climate means your pool's demands change significantly throughout the year. Adjusting your maintenance approach with the seasons prevents problems and saves money on chemicals and energy.

Summer (December – February)

Peak Season — Maximum Demand

Summer is when your pool works hardest. Higher temperatures accelerate chlorine consumption, UV intensity destroys sanitiser faster, and increased bather loads introduce more organic contaminants. Run the pump for **8–10 hours continuously**. Test chlorine and pH **once per week** (or more frequently during extreme heatwaves). Keep the pool covered when not in use to reduce evaporation and UV degradation. Maintain chlorine at the higher end of the range (approximately 4 ppm).

Autumn (March – May)

Transition Season — Leaf Management

Falling leaves are the primary challenge in autumn. Leaves that sink to the bottom decompose rapidly, consuming chlorine and feeding phosphate levels. Skim the surface daily, empty baskets frequently, and consider a leaf net over the pool if you have deciduous trees nearby. Reduce pump runtime to **6–8 hours**. This is an excellent time to have a professional service and equipment check before winter.

Winter (June – August)

Off-Season — Maintenance Mode

Do not turn your pool off for winter. A common mistake is to shut everything down, which leads to stagnant water, algae growth, and potential equipment damage from freezing in cold snaps. Reduce pump runtime to **4–6 hours**. Reduce chlorinator output to match the lower demand. Continue testing fortnightly at minimum. Keep the pool covered to prevent debris accumulation and heat loss. Maintain chemistry — it is far cheaper to maintain a clean pool through winter than to recover a green pool in spring.

Spring (September – November)

Opening Season — Preparation

Spring is the time to prepare for the swimming season. Remove and clean the pool cover. Increase pump runtime to **6–8 hours**. Perform a thorough backwash and consider a filter degreasing treatment. Do a comprehensive water test and rebalance all parameters. Brush the entire pool surface. Check all equipment for wear or damage after winter. Book a Palatial Pools spring service to ensure everything is in peak condition before the first swim.

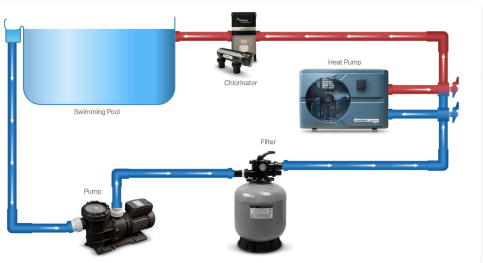
365 Days a Year

Your pool needs attention every season — even winter. Consistent care prevents costly recovery.

Heating & Energy Efficiency

Heating your pool extends the swimming season significantly and maximises the return on your investment. However, heating can be energy-intensive if not managed correctly. Choosing the right heating method and implementing efficiency measures can dramatically reduce running costs.

Heating Options



Inverter Heat Pumps (e.g., Neptune X-Pro)

Heat pumps extract ambient heat from the air and transfer it to the pool water — essentially acting as a reverse air conditioner. Modern **Full Inverter** heat pumps adjust their compressor speed based on demand, making them incredibly efficient. They are the most cost-effective way to maintain a constant warm temperature throughout the swimming season. The Neptune X-Pro range from Pool Pro Australia includes WiFi connectivity for remote monitoring and control.

Gas Heaters

Gas heaters burn natural gas or LPG to heat the water rapidly. They are ideal for **spas** or pools that only need to be heated occasionally for specific events, as they can raise the temperature much faster than a heat pump regardless of outside air temperature. However, running costs are significantly higher for sustained heating.

Solar Heating

Solar systems pump pool water through black capillary tubes on the roof, absorbing heat from the sun. While running costs are very low (only the cost of a small circulation pump), they are entirely dependent on sunny weather and provide no heating on overcast days or at night.

Maximising Efficiency

The Importance of Pool Covers

A pool cover is the **single most effective energy-saving device** you can own. Up to 70% of heat loss in a pool occurs through surface evaporation. A high-quality solar blanket (like the **Neptune SolarPro 525 Micron**) acts as a vapour barrier, drastically reducing heat loss overnight, cutting chemical consumption by preventing UV degradation and evaporation, and reducing your heating bill by up to 70%. If you heat your pool and do not use a cover, you are essentially heating the atmosphere.

Variable Speed Pumps

Upgrading to a variable speed pump (VSP) can reduce filtration energy costs by up to **80%**. Because water resistance decreases exponentially as flow rate drops (Affinity Laws), running a VSP at half speed uses only approximately one-eighth of the energy of a single-speed pump while still providing excellent filtration over a longer continuous runtime. Palatial Pools supplies and installs the Neptune X-Pro Full Inverter pump range.

Heating Method	Best For	Running Cost	Heating Speed
Inverter Heat Pump	Regular pool heating	Low–Medium	Moderate (12–48 hrs initial)
Gas Heater	Spas, occasional use	High	Fast (2–6 hrs)
Solar Heating	Summer extension	Very Low	Slow (weather dependent)
Solar Blanket	All pools (retention)	None	Passive heat gain

Source: Energy efficiency data referenced from Pool Pro Australia product specifications and SPASA Victoria guidelines.

Troubleshooting Common Issues

Even with diligent maintenance, occasional issues can arise. Here is a quick reference guide to diagnosing and resolving the most common pool problems.

Cloudy Water

Causes: Poor filtration (insufficient runtime or dirty filter), low chlorine, high pH reducing chlorine effectiveness, high Total Dissolved Solids (TDS), or the early stages of an algae bloom.

Solutions: Ensure the pump is running its full continuous block. Backwash the filter. Test and adjust pH to 7.2–7.4. Shock dose with liquid chlorine to raise free chlorine above 4 ppm. If the issue persists, use a clarifier or Pool Pro Pool Emeraldz phosphate remover.

Strong "Chlorine" Smell & Eye Irritation

Causes: Counterintuitively, a strong chemical smell means there is *not enough* free chlorine. The smell is caused by **chloramines** (combined chlorine) — the byproduct of chlorine reacting with sweat, urine, and body oils.

Solutions: The pool requires a "breakpoint chlorination" shock treatment. Add a large dose of liquid chlorine (typically 10x the combined chlorine reading) to destroy the chloramines and restore clean, odourless free chlorine.

Pump Has Lost Prime (No Water Flow)

Causes: Low pool water level, full skimmer/pump baskets, air leak in the suction line, cracked pump lid O-ring, or a stuck weir door in the skimmer box.

Solutions: Check the water level first. Empty all baskets. Inspect the pump lid O-ring for cracks or debris and ensure it is lubricated and seated correctly. Check all suction-side unions and valves for air leaks. If the pump runs dry for too long, it will overheat and melt the internal mechanical seal — an expensive repair.

Chlorinator "Low Salt" Warning

Causes: The salt level is genuinely low, the water is very cold (which lowers conductivity and gives a false low reading), or the salt cell is calcified and needs cleaning.

Solutions: Have the salt level independently tested with a digital salt meter before adding more salt — over-salting damages equipment and is difficult to reverse. Inspect the cell plates; if covered in white scale, clean the cell. If the cell is clean and salt levels are correct, the cell may be reaching end of life (typically 3–5 years).

Green or Discoloured Water

Causes: Algae bloom (green), dissolved metals such as copper or iron (green/brown), or tannin staining from leaves (brown/yellow).

Solutions: Test chlorine and pH immediately. If chlorine is low and pH is high, it is almost certainly algae — follow the Green Pool Recovery Protocol on the next page. If chlorine is adequate, the discolouration may be metal-based, requiring a metal sequestrant treatment rather than more chlorine.

> **When to Call Palatial Pools Immediately**
>
> Contact our service team if you experience: persistent water loss (dropping more than 5mm per day excluding evaporation), grinding or screeching noises from the pump, electrical faults or tripping safety switches, visible cracks in the pool shell, or any situation where you are unsure of the cause. **Turn off the equipment and call 0452 157 694.**

Green Pool Recovery Protocol

A pool turns green when algae blooms due to a sustained lack of effective sanitiser, poor circulation, or high phosphate levels providing a food source. Recovering a green pool requires aggressive chemical treatment, continuous filtration, and patience.

The 7-Step Recovery Process

1. **Remove Organic Debris:** Use a heavy-duty leaf shovel (like the Pool Pro Deluxe Leaf Shovel) to scoop out as much leaves and debris from the floor as possible. Chemicals cannot effectively fight algae if they are busy breaking down solid organic matter.
2. **Lower the pH to 7.0–7.2:** Algae treatments and high chlorine doses are far more effective at lower pH levels (remember — pH controls chlorine effectiveness). Use Hydrochloric Acid to bring the pH down before shocking.
3. **Shock the Pool Aggressively:** Add a massive dose of Pool Pro Liquid Pool Chlorine. A standard 50,000L pool with a severe bloom may require 15–20 litres of liquid chlorine. The goal is to achieve a free chlorine reading above 10 ppm.
4. **Add Algaecide:** Follow up with a high-strength copper-based or polyquat algaecide to break down the algae's protective outer cell walls, making it more vulnerable to chlorine.
5. **Brush Vigorously:** Brush the entire pool surface — walls, floor, steps, and behind ladders. Algae clings to surfaces and forms a protective biofilm that shields it from chemicals. Brushing exposes it.
6. **Run Pump 24/7:** Do not turn the pump off until the pool is completely clear. The dead algae must be filtered out of the water continuously.
7. **Backwash Frequently:** As the filter catches the dead algae, pressure will rise rapidly. You may need to backwash daily, or even twice a day, during the recovery process.

The Final Polish: Phosphate Removal

Phosphates are algae food. Once the pool is clear, test for phosphates. If levels are elevated, use a phosphate remover (like **Pool Pro Pool Emeraldz**) to starve any remaining algae spores and prevent the bloom from returning. Keeping phosphates below 100 ppb significantly reduces the risk of future algae outbreaks.

What NOT to Do

- Do NOT add flocculant to a green pool and then try to vacuum — this often makes the situation worse if the filter cannot handle the load.
- Do NOT drain the pool without professional advice. Draining can cause a fibreglass shell to float or a concrete shell to crack from groundwater pressure.
- Do NOT add stabilised chlorine (tablets/granules) for shock treatment — they add cyanuric acid, which compounds the problem.

If the pool is "swamp green" and you cannot see the bottom, it may be more cost-effective and faster to have Palatial Pools perform a professional **Green Pool Recovery** service, which may involve flocculants, specialised vacuuming to waste, or in extreme cases, draining and acid washing the shell.

Prevention Is Always Cheaper Than Recovery

A green pool recovery can cost hundreds of dollars in chemicals alone, plus days of intensive filtration and labour. By contrast, maintaining correct chemistry through simple weekly testing of chlorine and pH costs very little and takes only a few minutes. The single most effective way to prevent a green pool is to **never let the chlorine drop below 2 ppm** and **never let the pH rise above 7.6**. If you are going on holiday, consider asking a neighbour to check the pool or book a Palatial Pools maintenance visit to cover the period.

Quick Reference: Green Pool Warning Signs

Stage 1 — Dull/Hazy Water: The water loses its sparkle and appears slightly cloudy. Chlorine is likely low. Test and dose immediately.

Stage 2 — Light Green Tint: Algae is beginning to bloom. Shock dose with liquid chlorine, lower pH to 7.0–7.2, and run the pump 24/7.

Stage 3 — Solid Green/Cannot See Bottom: Full algae bloom. Follow the 7-Step Recovery Protocol above or call Palatial Pools for professional recovery.

Melbourne Seasonal Pool Care Timeline

SPRING (Sep–Nov) — Opening Season

Book professional start-up service • Balance chemistry • Inspect equipment • Increase pump runtime to 6–8 hrs • Check salt cell • Clean filter

SUMMER (Dec–Feb) — Peak Season

Test chlorine & pH weekly • Run pump 8–10 hrs daily • Backwash as needed • Skim debris daily • Monitor chlorine demand after heavy use • Keep chlorine at ~4 ppm

AUTUMN (Mar–May) — Winding Down

Reduce pump runtime to 4–6 hrs • Lower chlorinator output • Treat phosphates • Clean filter thoroughly • Clear all baskets • Book pre-winter service

WINTER (Jun–Aug) — Maintenance Mode

Run pump 3–4 hrs daily • Maintain minimum 2 ppm chlorine • Test monthly • Remove leaves promptly • Check water level • Do NOT neglect the pool entirely

Source: Adapted from SPASA Victoria seasonal pool care guidelines.

Pool Owner's Emergency Quick Reference

Situation	Immediate Action
Pool turning green	Lower pH to 7.0–7.2, shock dose with liquid chlorine, run pump 24/7
Pump not priming	Check water level, empty baskets, inspect lid O-ring, check for air leaks
Chemical spill	Contain with sand/soil, do NOT wash to stormwater, call 000 if mixed chemicals
Suspected leak	Mark water level, wait 24 hrs (pump off), if >5mm loss call Palatial Pools
Power outage	Manually dose liquid chlorine daily until pump resumes, cover pool if possible

Palatial Pools Emergency Line: 0426 215 215

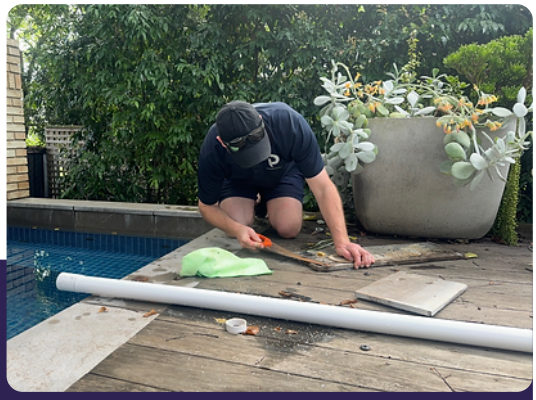
Palatial Pool Services

Palatial Pools is a premier boutique pool and spa service based in Melbourne, renowned for delivering unparalleled expertise and exceptional customer service at fair and transparent prices. We are proud to be the **#1 Rated Pool Service in the City of Yarra**, backed by a 10-year workmanship guarantee on our installations and complex repairs.

Our Core Services

Regular Maintenance & Servicing

Take the hassle out of pool ownership with our scheduled maintenance programs. We handle the chemistry testing and adjustment, equipment checks, cleaning, and preventive maintenance so you can simply enjoy the water.



Regular Monthly Servicing

$150 per month

Comprehensive scheduled maintenance including chemistry balancing, equipment inspection, basket clearing, and system optimisation.

Single Visit Service

$160 per visit

One-off service call including 1 hour of labour. Ideal for troubleshooting, seasonal check-ups, or getting a second opinion.

Licensed Plumbing

Comprehensive swimming pool plumbing including new pool fitouts, renovations, equipment relocations, underground leak repairs, and pipe replacements. We also complete certified, licensed plumbing operations such as sewage connections for backwash lines where required by council.

In-House Electrical

Our in-house certified electrician specialises in pool equipment connections — heat pump installations, outlet additions, lighting upgrades, and electrical fault diagnosis. Having electrical services in-house means faster turnaround and better coordination with our plumbing and equipment work.

Gas Heater Services

In-house certified gas pool services including heater connections, gas leak detection, gas plumbing, new valve installations, pressure testing, and diagnostics. Affordable and professional with decades of experience.

Equipment Supply & Installation

We supply and install a full range of pool equipment from trusted Australian brands. Whether you need a new pump, filter, chlorinator, heat pump, or automation controller, we can source the right unit for your pool size and budget, and install it to manufacturer specifications. We only recommend equipment we would use on our own pools — no upselling, no unnecessary upgrades.

Pool Renovations & Resurfacing

From minor tile repairs to full interior resurfacing, Palatial Pools can restore your pool to its original condition or give it a completely new look. We work with pebblecrete, glass bead, quartzon, and tiled finishes. Our renovation projects include coping replacement, waterline tile upgrades, and structural crack repairs.

10-Year Workmanship Guarantee

We stand by the quality of our installations and complex repairs. Palatial Pools offers a **10-year workmanship guarantee** on our plumbing and installation services, providing you with absolute peace of mind that the job has been done right.

Why Choose Palatial Pools?

We are a small, owner-operated business that treats every pool as if it were our own. We do not employ subcontractors — every job is completed by our qualified in-house team. This means consistent quality, direct communication, and genuine accountability. Our clients stay with us because we give honest advice, charge fair prices, and stand behind our work with a 10-year guarantee.

Seasonal Service Reminders

Spring Opening (September–October): Before the swimming season begins, book a professional start-up service. We inspect all equipment, prime the pump, check for winter damage, balance the water chemistry, and ensure the chlorinator is producing correctly. Starting the season with properly balanced water prevents problems all summer.

Autumn Closing (April–May): As temperatures drop, reduce pump runtime gradually and lower chlorinator output. Clean the filter, clear all baskets, and consider a phosphate treatment to prevent algae establishing over winter. A pre-winter service ensures your pool stays manageable through the colder months with minimal intervention.



Advanced Diagnostics & Leak Detection

Palatial Pools has invested in the latest and most accurate diagnostic and leak detection devices currently available. Our advanced technology suite allows us to find problems that other pool companies simply cannot — saving you time, money, and unnecessary excavation.

Thermal Imaging



Our high-precision **thermal imaging cameras** detect temperature differentials in pool surrounds, decking, and equipment areas that are invisible to the naked eye. When water leaks from underground pipes, it creates a temperature signature on the surface above. Thermal imaging allows us to pinpoint the location of a leak without any digging, dramatically reducing the cost and disruption of repairs.

Thermal imaging is also invaluable for diagnosing heat loss issues, identifying blocked pipes (which show different thermal signatures), and verifying the integrity of repairs after completion.

Acoustic Leak Detection



Our **underground acoustic mapping equipment** uses highly sensitive microphones and ground sensors to listen for the sound of water escaping from pressurised pipes beneath the pool deck and surrounding areas. Even tiny leaks produce a characteristic sound signature that our equipment can detect and localise with remarkable precision.

By combining acoustic detection with thermal imaging, we can cross-reference findings and confirm leak locations with a high degree of confidence before any excavation begins. This dual-technology approach is what sets Palatial Pools apart from competitors who rely on guesswork or destructive trial-and-error methods.

Acoustic & Infrared Leak Detection

From $345

Includes 1.5 hours of on-site labour with full diagnostic report. Additional time billed at standard rates if required.

Additional Diagnostic Services

Service	Description
Pressure Testing	Pressurising individual pipe runs to identify which line is leaking and quantify the leak rate.
Dye Testing	Using specialised dye near suspected leak points to visually confirm water movement through cracks or joints.
Electronic Leak Detection	Using electrical conductivity methods to locate breaches in vinyl and fibreglass pool shells.
Camera Inspection	Inserting a waterproof camera into pipes to visually inspect for cracks, root intrusion, or blockages.

Onsite Equipment Lessons

Palatial Pools can provide **onsite equipment lessons** by our qualified Trainer and Assessor (TAE10). Whether you are a new pool owner, have just moved into a property with an unfamiliar system, or simply want a refresher course, we can teach you about all aspects of your individual pool — including chemistry, equipment operation, backwashing, and your responsibilities as a pool owner. Learn how to maintain your pool or spa the right way!

Source: Leak detection methodology referenced from Palatial Pool Services (www.palatialpools.com.au).

Safety & Compliance (Victoria)

Pool safety is a critical legal responsibility for all pool and spa owners in Victoria. The regulations are designed to prevent tragic accidents, particularly involving young children. Non-compliance can result in significant fines and, more importantly, puts lives at risk.

Victorian Pool Barrier Laws

In Victoria, strict laws govern swimming pool and spa barriers. All pools and spas capable of holding more than 300mm (30cm) of water must have a compliant safety barrier under the **Building Regulations 2018** and **Australian Standard AS 1926.1**.

Key Requirements

- **Registration:** All pools and spas must be registered with your local council. Failure to register can result in fines.
- **Barrier Inspection:** Barriers must be inspected by a registered building surveyor or qualified pool barrier inspector every **four years**. A certificate of barrier compliance must be lodged with the council.

- **Gate Standards:** Pool gates must be self-closing and self-latching from any position. The latch must be at least 1500mm from ground level (or 1200mm if on the inside with a specific shield). Gates must **never** be propped open.
- **Barrier Height:** The barrier must be at least 1200mm high with no gaps greater than 100mm.
- **Non-Climbable Zone (NCZ):** A 900mm NCZ must be maintained on the outside of the barrier. Trees, BBQs, pot plants, chairs, and outdoor furniture must be kept away from the fence to prevent children from using them to climb over.
- **Window Compliance:** Any window within the barrier that opens must have a compliant restrictor or be treated as part of the barrier.

> **Active Supervision is Essential**
>
> A compliant barrier is a legal requirement, but it is **not a substitute for active adult supervision**. Never assume a child is safe because the pool is fenced or because they are wearing flotation devices. Drowning can occur silently in seconds. Designate a responsible adult as the "water watcher" whenever children are near the pool — this person should not be distracted by phones, conversations, or other activities.

Chemical Handling Compliance

Pool chemicals are classified as hazardous substances under **WorkSafe Victoria** regulations. Even for residential use, safe handling practices are essential.

- Always wear appropriate PPE (safety glasses and chemical-resistant gloves) when handling acids or concentrated chlorine.
- Ensure the storage area is well-ventilated to prevent toxic fume buildup.
- Keep Safety Data Sheets (SDS) accessible for all chemicals stored on your property. These are available from Pool Pro Australia's website.
- In the event of a chemical spill, contain it with sand or soil. Do **not** wash chemicals into the stormwater drain. Contact emergency services if the spill involves mixed reactive chemicals.

Source: Victorian Building Authority (VBA), Building Regulations 2018, AS 1926.1-2012, and WorkSafe Victoria.

Signs Your Pool Needs Immediate Attention

GREEN OR CLOUDY WATER — Chlorine is likely depleted. Test immediately, shock dose, and run the pump 24/7 until clear.

STRONG CHLORINE SMELL — Paradoxically means chlorine is too LOW, not too high. Combined chloramines (chlorine + organic waste) cause the smell. Shock the pool.

WATER LEVEL DROPPING FAST — If losing more than 5mm/day (beyond normal evaporation), suspect a leak. Call Palatial Pools for diagnostic testing.

PUMP MAKING UNUSUAL NOISES — Grinding, screeching, or cavitation sounds indicate bearing failure, air leaks, or blockages. Turn off and call for service.

FILTER PRESSURE CONSISTENTLY HIGH — If pressure stays high after backwashing, the media may be exhausted or channelled. Time for a filter service or media replacement.

Your Pool Details

Keep this page updated with your specific pool information. Having these details on hand makes ordering parts, calculating chemical doses, and requesting service much easier and faster.

Pool Specifications

Pool Volume (Litres): _____

Surface Type: _____

Sanitiser Type: _____

Pool Shape & Dimensions: _____

Equipment Log

Pool Pump Make/Model: _____

Filter Make/Model: _____

Filter Media Type: _____

Chlorinator Make/Model: _____

Heater Make/Model: _____

Automation/Controller: _____

Service History

Last Filter Media Change: _____

Last Professional Service: _____

Last Salt Cell Replacement: _____

Barrier Inspection Due Date: _____

Calculating Pool Volume

Rectangular: Length (m) × Width (m) × Average Depth (m) × 1,000 = Litres

Circular: Diameter (m) × Diameter (m) × Average Depth (m) × 785 = Litres

Kidney/Freeform: Length (m) × Width (m) × Average Depth (m) × 750 = Litres (approximate)

Example: 8m × 4m × 1.5m × 1,000 = 48,000 Litres

Notes

We're Here to Help

Thank you for choosing Palatial Pool Services. If you ever need assistance, parts, or professional servicing, please reach out to our team.

Phone: 0452 157 694

Email: service@palatialpools.com.au

Web: www.palatialpools.com.au

Mon–Fri 9am–6pm • Sat 9am–5pm • Melbourne Pools, Made Easy

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